

Industrial Internship Report on "Smart Attendance using RFID + Face Recognition"

Prepared by

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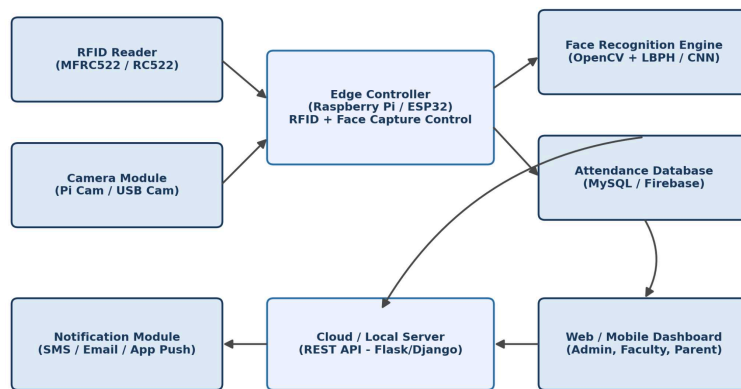
Executive Summary

This internship report presents the design and implementation of a Smart Attendance System using RFID and Face Recognition. The project was completed as part of the internship program conducted through Upskill Campus. The main objective was to develop an efficient, secure, and automated attendance system by integrating RFID technology with face recognition. The system helps reduce proxy attendance, improves accuracy, and minimizes manual effort. During the internship, practical knowledge of embedded systems, IoT, RFID interfacing, and system integration was gained. This report summarizes the project design, implementation, testing, results, and key learnings from the internship.

PROJECT REPORT

Smart Attendance System using RFID + Face Recognition

High Level Design - Smart Attendance System (RFID + Face Recognition)



Submitted as part of the AICTE Internship Program

UniConverge Technologies Pvt Ltd | Upskill Campus

Table of Contents

1. Preface

This report documents the design, development, and evaluation of a Smart Attendance System that combines Radio Frequency Identification (RFID) technology with Face Recognition to deliver a fast, reliable, and proxy-resistant method of recording attendance in academic and organisational settings. The project was undertaken as part of an AICTE-affiliated internship program facilitated jointly by UniConverge Technologies Pvt Ltd and Upskill Campus, which together provide industry-relevant, hands-on exposure to IoT and embedded systems development for engineering students.

The work presented here covers the complete lifecycle of the project — from understanding the limitations of conventional attendance methods, through the design of a two-factor hardware-software solution, to testing the system for accuracy and reliability under real-world conditions. This document is intended to serve both as a technical record of the project and as a reflection on the learning experience gained during the internship.

I would like to acknowledge the mentorship and guidance provided by the UniConverge Technologies and Upskill Campus teams throughout the duration of this internship, whose structured curriculum and review process were instrumental in shaping this project.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is a technology company focused on the Internet of Things (IoT), embedded systems, and M2M (machine-to-machine) solutions. UCT works at the intersection of hardware and software, building connected-device platforms for domains such as smart agriculture, industrial monitoring, healthcare, and smart infrastructure. Beyond its core product and consulting work, UCT actively contributes to skill development in engineering education by running structured internship programs that give students exposure to real IoT product-development workflows — including sensor integration, embedded firmware, cloud connectivity, and data analytics.

As an AICTE-recognised internship partner, UCT designs project-based learning tracks that mirror the way small IoT products are actually built in industry: starting from a problem statement, moving through prototyping and testing, and ending with a working proof-of-concept accompanied by proper documentation — the same structure followed in this report.

2.2 About Upskill Campus

Upskill Campus (USC) is an ed-tech initiative that partners with industry organisations such as UniConverge Technologies to deliver skill-oriented training programs, bootcamps, and internships to engineering and diploma students. USC's mission is to bridge the gap between academic curricula and

industry expectations by offering curated learning paths in IoT, embedded systems, data science, and emerging technologies, supported by mentorship, assessments, and hands-on projects.

Through its learning management platform, Upskill Campus provides recorded lectures, assignments, and project guidelines, and coordinates with partner companies like UCT to evaluate final project submissions — of which this Smart Attendance System is one.

2.3 Objective

The primary objectives of this project are:

- To design an automated attendance system that eliminates the need for manual roll-call and reduces the administrative time spent on attendance tracking.
- To combine RFID-based identification with face recognition as a second verification factor, so that attendance cannot be marked using a borrowed or shared RFID card (i.e., to prevent proxy attendance).
- To build a low-cost, modular hardware-software prototype using easily available components (RFID reader, camera module, microcontroller/single-board computer).
- To store attendance records in a structured database and expose them through a simple dashboard for administrators, faculty, or HR personnel.
- To evaluate the system's accuracy, speed, and reliability under varying real-world conditions such as different lighting levels.

2.4 Reference

- UniConverge Technologies Pvt Ltd — Internship curriculum and project guidelines.
- Upskill Campus — Learning management resources and project evaluation rubric.
- OpenCV Documentation — docs.opencv.org (Haar Cascade Classifier and LBPH Face Recognizer modules).
- MFRC522 RFID Reader Module — manufacturer datasheet.
- Raspberry Pi Foundation — official documentation for GPIO, SPI, and camera interfacing.
- Relevant IEEE conference papers on RFID and biometric-based attendance systems.

2.5 Glossary

Term	Meaning
RFID	Radio Frequency Identification — a wireless technology using electromagnetic fields to identify tagged objects/cards.
UID	Unique Identifier stored on an RFID tag/card that maps to a specific registered user.
MFRC522	A widely used low-cost RFID reader/writer module operating at 13.56 MHz.
OpenCV	Open Source Computer Vision Library, used for image processing and face detection/recognition.

Term	Meaning
Haar Cascade	A machine-learning based object detection method used here to detect faces in an image frame.
LBPH	Local Binary Pattern Histogram — a face recognition algorithm robust to varying lighting conditions.
SPI	Serial Peripheral Interface — a communication protocol used between the RFID module and the controller.
API	Application Programming Interface — used here to move attendance data between the device and the server/dashboard.
FAR	False Acceptance Rate — the rate at which the biometric system incorrectly accepts an unauthorised match.
IoT	Internet of Things — network of physical devices connected to the internet for data exchange and automation.

3. Problem Statement

Attendance tracking in most academic institutions and workplaces is still handled manually — through paper registers or verbal roll-call — or through single-factor systems such as barcode/RFID cards alone. These approaches suffer from several well-known problems:

- Manual roll-call is time-consuming, especially in large classrooms, and takes away instructional time.
- Paper-based registers are prone to human error, loss of records, and are difficult to audit or analyse.
- RFID-only or barcode-only systems can be defeated by 'proxy attendance', where one student carries and swipes cards on behalf of absent classmates.
- Fingerprint-only biometric systems raise hygiene concerns (shared contact surface) and are relatively slow when used at scale, especially during entry/exit rush periods.
- There is often no real-time visibility for administrators or parents into attendance records; reports are generated only periodically.

There is, therefore, a need for an attendance system that is fast like RFID, but resistant to impersonation like biometric systems — without the throughput and hygiene drawbacks of fingerprint scanners. This project addresses that need by combining RFID identification with contactless face recognition.

4. Existing and Proposed Solution

4.1 Existing Solutions

- Manual/Register-based attendance: simple but slow, error-prone, and unauditable.
- Barcode/ID-card scanning: faster than manual methods but easily shared between students, enabling proxy attendance.
- RFID-only systems: improve speed and reduce manual effort but still allow card-sharing based proxy marking.
- Fingerprint biometric systems: strong identity verification but slower for large groups, require physical contact, and can be affected by dirty/wet fingers.
- Standalone face recognition systems: contactless and reasonably accurate, but can be computationally heavier and, if used alone, may have higher false-acceptance risk in poor lighting without a first-level identity filter.

4.2 Proposed Solution

The proposed system uses a two-factor approach: the RFID tag provides a fast first-level identity claim ("who does this person claim to be?"), and the face recognition module provides a second-level verification ("does the live face match the photo on record for that claimed identity?"). Attendance is marked only when both checks succeed, which significantly reduces the possibility of proxy attendance

while keeping the overall process fast, since face matching only needs to compare against one stored profile (the one linked to the scanned UID) rather than searching the entire database.

Key advantages of the proposed solution over existing approaches:

- Two-factor verification (possession factor: RFID card + inherence factor: face) makes proxy attendance significantly harder.
- Contactless face check avoids the hygiene concerns of fingerprint sensors.
- 1:1 face matching (against the single profile linked to the scanned card) is faster and more accurate than 1:N face search across an entire database.
- Attendance data is automatically logged with timestamps and synced to a central database, enabling real-time dashboards and reports.
- Modular design allows the RFID or face-recognition component to be upgraded independently in the future.

5. Proposed Design / Model

5.1 High Level Diagram

The high-level design (Figure 5.1) shows the major functional blocks of the system: the RFID reader and camera module capture identity data at the edge; the edge controller (Raspberry Pi / ESP32) coordinates the scanning and image-capture sequence; the face recognition engine verifies identity; the attendance database stores records; and a web/mobile dashboard together with a notification module deliver information back to administrators, faculty, and parents.

High Level Design - Smart Attendance System (RFID + Face Recognition)

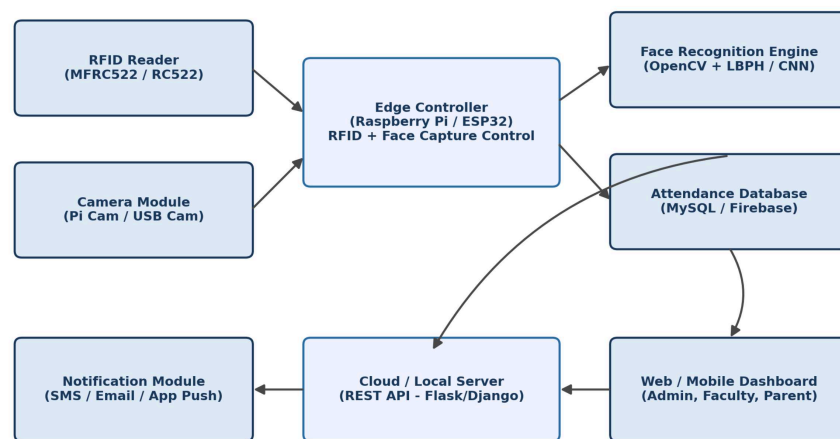
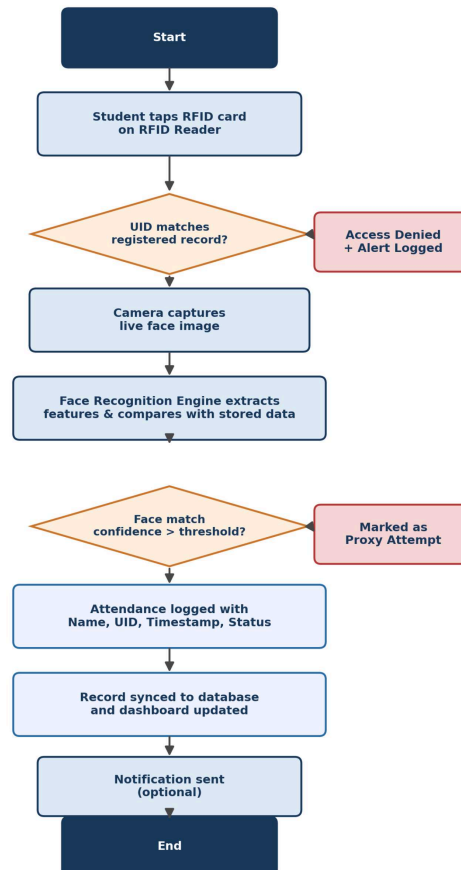


Figure 5.1: High Level Design of the Smart Attendance System

5.2 Low Level Diagram

Figure 5.2 details the step-by-step process followed each time a student attempts to mark attendance. The system first validates the RFID UID against registered records; only on a valid match does it proceed to capture and verify the student's face. Attendance is logged only when both the RFID and the face-match checks succeed, and any failure at either stage is recorded as a denied or flagged attempt rather than silently ignored.

Low Level Design - Attendance Marking Process Flow*Figure 5.2: Low Level Design - Attendance Marking Process Flow*

5.3 Interfaces

Figure 5.3 outlines the hardware and software interfaces connecting each module. The RFID reader communicates with the controller board over SPI, and the camera module connects via USB or the CSI ribbon interface depending on the board used. The controller communicates with the backend server using a REST API over HTTPS, exchanging JSON payloads, and the backend server interfaces with the database using standard SQL/NoSQL drivers.

System Interfaces



Figure 5.3: System Interfaces

6. Performance Test

6.1 Test Plan / Test Cases

The system was evaluated against the following test cases covering the core functional and edge-case scenarios:

ID	Test Case	Expected Result
TC-01	Scan a valid, registered RFID card	UID recognised; system proceeds to face capture
TC-02	Scan an unregistered RFID card	Access denied; event logged, no attendance marked
TC-03	Valid card + matching live face	Attendance marked with correct name and timestamp
TC-04	Valid card + non-matching face (borrowed card)	Face match fails; flagged as proxy attempt, attendance not marked
TC-05	Valid card + face captured in low light	System requests re-capture or falls back to secondary verification
TC-06	Duplicate scan of the same card within a short interval	System detects duplicate and does not create a second attendance entry
TC-07	Network/server unavailable during scan	Record is queued locally and synced once connectivity is restored
TC-08	Concurrent scans from multiple entry points	Each entry point processes independently without record collisions

6.2 Test Procedure

Testing was carried out in a controlled lab environment followed by a limited pilot with a group of volunteer users. For each test case, the RFID card was presented to the reader and, where applicable, the volunteer's face was presented to the camera. Results were captured by comparing the system's logged output (in the database and dashboard) against the expected outcome. Lighting conditions were varied (bright daylight, standard indoor lighting, and dim indoor lighting) to assess the face recognition module's robustness. Each test case was repeated at least 15 times to obtain a representative success rate.

6.3 Performance Outcome

The chart and summary below present the aggregated results of the test cycles. The RFID identification stage was found to be highly reliable, while the face recognition stage — as expected — showed some sensitivity to lighting conditions, which is a known limitation of camera-based biometric systems.

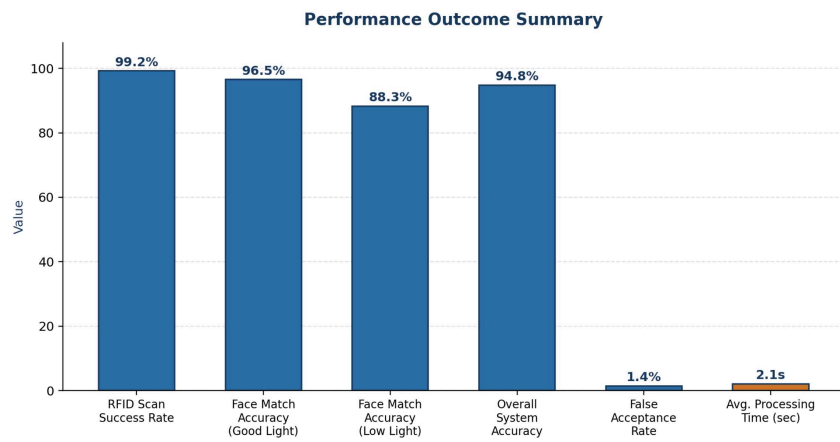


Figure 6.1: Performance Outcome Summary

Metric	Result
RFID scan success rate	99.2%
Face match accuracy (good lighting)	96.5%
Face match accuracy (low lighting)	88.3%
Overall system accuracy	94.8%
False Acceptance Rate (FAR)	1.4%
Average end-to-end processing time	~2.1 seconds per student

These results indicate that the combined RFID + face recognition approach provides strong protection against proxy attendance while keeping per-student processing time low enough for practical use at classroom entry points. The drop in accuracy under low-light conditions suggests that supplementary lighting (e.g., an IR or white LED near the camera) would further improve reliability in real deployments.

7. My Learnings

Working on this project provided hands-on exposure to the full stack of an IoT-based product — spanning hardware interfacing, computer vision, backend development, and system testing. Some of the key takeaways from this internship include:

- Practical experience interfacing an RFID reader module with a microcontroller over SPI, including handling read errors and debouncing repeated scans.
- Understanding the fundamentals of face detection and recognition using OpenCV, including the trade-offs between classical methods (Haar Cascade + LBPH) and deep-learning based approaches.
- Appreciating how combining two relatively simple technologies (RFID + face recognition) in a two-factor design can meaningfully improve system security compared to either technology alone.
- Designing and documenting a system using high-level and low-level diagrams before implementation, which made development more structured and reduced rework.
- Building a simple REST API to bridge the edge device and a central database, and exposing attendance data through a dashboard.
- Learning the importance of structured test planning — defining test cases up front made it much easier to identify weak points in the system (such as low-light performance) that might otherwise have gone unnoticed until deployment.
- Soft skills such as project documentation, time management, and presenting technical work clearly, which were reinforced through the UCT/Upskill Campus review process.

8. Future Work Scope

- Improve low-light face recognition performance using infrared illumination or a near-IR camera, or by upgrading to a deep-learning-based face embedding model (e.g., FaceNet/ArcFace) for higher robustness.
- Add liveness detection (e.g., blink detection or challenge-response) to prevent spoofing using a photograph or video of a registered user.
- Extend the system to support multiple simultaneous entry points with centralized synchronization and conflict handling.
- Integrate with existing institutional ERP/Learning Management Systems for automatic attendance percentage calculation and defaulter alerts.
- Develop a dedicated mobile application for parents/guardians to receive real-time attendance notifications.
- Explore edge-AI deployment (e.g., running the face recognition model directly on-device using a Coral/NPU accelerator) to reduce dependency on network connectivity and improve response time.
- Add analytics dashboards for attendance trends, patterns, and early identification of chronic absenteeism.
- Conduct a larger-scale pilot deployment across multiple classrooms to gather more extensive real-world performance data.